# Table of contents

Introduction . . . . . 1
Line breaking . . . . . 1
Hyphenation . . . . . 2
Floats and inserts . . . . . 2
Closing remarks . . . . . 2

## Introduction

This document demonstrates automatic cross-references with page numbers. The table of contents above is not filled in by hand. The numbers after each link are rendered in CSS via `target-counter(attr(href), page)`.

The first glu pass does not yet know the page numbers and renders `?`. The next pass, after the anchor positions have been written to the aux file, has the real values. glu's multi-pass loop converges automatically.

## Line breaking

The TeX algorithm by Knuth and Plass minimises global badness instead of breaking greedily line by line. boxes and glue ports it to Go.

A paragraph turns into a list of glyph, box, and glue nodes. The solver picks the decomposition with the lowest total cost, which means fewer orphans and widows than greedy CSS line breaking gives you.

## Hyphenation

The hyphenation patterns follow Liang/Knuth. CSS Text 3 `hyphens: auto` and `hyphens: manual` are supported, and soft-hyphens (``) are honoured.

## Floats and inserts

Floats can be anchored to the top or bottom of a page via the CSS `float` property. Footnotes use the same insert mechanism but live in their own class with a separator rule.

## Closing remarks

The table of contents at the top of this document is the demonstration of `target-counter()`. Try what happens when you add or remove a section: all following page numbers update automatically as soon as glu runs its second pass.